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Procedia Computer Science 251 (2024) 334–341

Procedia Computer Science

www.elsevier.com/locate/procedia

The 14th International Conference on Current and Future Trends of Information and Communication Technologies in Healthcare (ICTH 2024)
October 28-30, 2024, Leuven, Belgium

SentimentCareBot: Retrieval-Augmented Generation Chatbot for Mental Health Support with Sentiment Analysis

Jean Pierre Nayinzira[a,*], Mehdi Adda[a]

[a]*University of Quebec at Rimouski, 300 allée des Ursulines, Rimouski G5L 3A1, Canada*

Abstract

The global mental healthcare system faces various challenges in terms of accessibility and the availability of specialist support, such as psychologists and counselors, especially following the COVID-19 pandemic. This study explores a potential solution to this problem by developing a chatbot model, SentimentCareBot, which integrates sentiment analysis with retrieved-augmented generation (RAG) techniques and Large Language Models (LLMs). The study uses a public Mental Health Counseling Conversations Dataset and baseline selection methods such as Naive RAG, Multi-query RAG, and Hypothetical Document Embeddings (HyDE) to improve query translations. The findings from Tukey's Honest Significant Difference (HSD) test reveals a significant improvement in sentiment analysis performance when it is applied to the Multi-query RAG using the MistralAI language model, compared to both Multi-query RAG using the OpenAI language model and HyDE using OpenAI with Sentiment Analysis. These results demonstrate the potential of sentiment analysis to enhance the effectiveness of mental health chatbots.

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Peer-review under responsibility of the scientific committee of the Conference Program Chairs

Keywords: Mental Health; Chatbot; Retrieval-Augmented Generation (RAG); Sentiment Analysis; Large Language Models (LLMs).

1. Introduction

The World Health Organization (WHO) brings attention to the critical importance of mental health as a fundamental human well-being, by promoting the improvement of global mental health. They focus on the significant gap between the demand for mental health services and their current availability, despite different public health crises, in particular the COVID-19 pandemic. WHO recommends integrating mental health services into primary healthcare and achieving universal health coverage [24]. In addition, the WHO points out the essential role of healthcare professionals in expanding mental healthcare, recommending increased training and support networks to improve service provision. It calls on governments to boost mental health funding to facilitate these necessary changes.

* Corresponding author. Tel.: +1-418-509-7757.
E-mail address: JeanPierre.Nayinzira@uqar.ca

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10.1016/j.procs.2024.11.118

In a related study, Abd-Alrazaq et al. [1] explore the potential of chatbots in mental healthcare, noting their effectiveness in addressing disorders such as depression and stress. Although acknowledging the need for improvements in chatbot conversation capabilities and integration into healthcare, the study highlights patient perceptions and opinions, pointing to chatbots as a cost-effective and accessible complement to traditional mental health treatments.

While chatbots have proven effective in mental health contexts, several challenges remain. One key issue is the ethical concern about the reliability and effectiveness of these chatbots, many of which lack evidence-based support or sufficient research backing. It is crucial for the development and deployment of mental health chatbots to be anchored in clinical evidence to confirm their effectiveness [10]. In addition, a significant limitation of chatbots in mental health care is their fundamental inability to experience and convey empathy as humans do. Although some users find chatbots, such as Woebot, to be empathetic and supportive, this perception is not universal. The experience of empathy from chatbots can differ significantly among individuals [5].

In this study, we introduce a chatbot named "SentimentCareBot" by implementing Retrieval-Augmented Generation (RAG) models across different baseline selections (section 3.4) and then applying sentiment analysis to all of these models. To evaluate these models, we used two datasets: a complete evaluation dataset comprising 106 question-answer pairs, and a subset of this dataset consisting of five question-answer pairs. For evaluation, we used Ragas evaluation metrics to calculate *Faithfulness*, *Answer Relevancy*, and *Answer Correctness* scores for the various RAG models. Additionally, we selected two Large Language Models (LLMs) for our evaluation: OpenAI's "gpt-3.5-turbo-0125" [23] and Mistral's "mistral-large-latest" [21]. Finally, we performed an ANOVA and Tukey HSD test to analyze the significant differences between various RAG models, applying sentiment analysis to each scenario. Our results also included an analysis of token usage and latency for both LLMs, using the subset evaluation data to demonstrate the API's limitations on the study.

2. Background

With the proposal of the Turing test by Alan Turing in 1950, which posed the question "Can machines think?", the foundation for chatbot development was laid, signaling the start of their popularization. The first notable chatbot, Eliza, developed in 1966, mimicked a psychotherapist by employing simple pattern matching and template-based responses to echo user input as questions. Despite its rudimentary conversational abilities, Eliza managed to confuse people during an era when computer interaction was rare, thus encouraging the creation of more sophisticated chatbots. Significant advancements was achieved with the introduction of PARRY in 1972, a chatbot designed with a distinct personality. Later, the 1995 creation of the chatbot ALICE, which was awarded the "most human computer" title after winning the Loebner Prize in 2000, 2001, and 2004, showcased further progress. ALICE used a pattern matching algorithm enhanced by the Artificial Intelligence Markup Language (AIML), facilitating the expansion of its knowledge base by developers [2].

Machine learning and natural language processing advancements have facilitated the development of advanced chatbots, including Amazon's Alexa and Google's Assistant, reducing reliance on rules and pattern matching. These innovations have improved chatbot flexibility, ease of implementation, and the precision of human-like conversation, presenting clear advantages over rule-based models in terms of adaptability and domain independence. Furthermore, machine learning enables chatbots to learn directly from human dialogue, eliminating the need to manually define patterns, and thus increasing their adaptability [6].

Within the framework of Seq2Seq (Sequence-to-Sequence) models, the attention mechanism plays a crucial role in overcoming the limitations posed by encoding entire input sentences into fixed-length context vectors. This innovative mechanism enables the decoder to selectively focus on various parts of the input sequence, effectively preserving vital information and context that could be lost in longer sequences. By forming direct connections between the target and source, the attention mechanism allows the model to concentrate on relevant segments of the input during translation or response generation, thereby enhancing the model's capability to handle variable-length sequences and improving the simplicity and clarity of generated responses. Such advancements make the attention mechanism a crucial element in the development of more sophisticated and human-like conversational agents.

Further insights on the evolution of chatbot technologies are transformer models that employ multiple attention mechanisms and demonstrate better performance and accuracy over traditional Seq2Seq models equipped with attention mechanisms. The key to the transformer model's success lies in its ability to address sequence-related challenges

without relying on recurrent neural networks (RNNs), thus optimizing training time and enhancing neural machine translation performance. Nuanced handling and interpretation of input sequences facilitated by the attention mechanisms within transformers result in more accurate and contextually relevant responses. This leap forward marks a significant step in the ongoing effort to harness advanced neural network techniques, including Long-Short-Term Memory (LSTM) cells [20, 25].

The Transformer Architecture, which employs a self-attention mechanism to process complex data sequences, is central to the development of LLMs. This design facilitates parallel computations and effectively manages long-distance dependencies in text, laying the groundwork for the Generative Pre-trained Transformer (GPT) series. Notably, these models excel in generating text that mirrors human quality and achieve high accuracy in various linguistic tasks, thanks to extensive training on broad textual corpora [8].

Despite these technological strides, integrating LLMs into healthcare presents distinct challenges, particularly medical misinformation. Concerns include the misinterpretation of medical terms and the generation of advice that contradicts standard medical practices. Furthermore, the use of commercial servers by LLMs to store patient health data has sparked debates about privacy implications [26].

Addressing the limitations of LLMs, the RAG framework offers a distinct approach by tackling the challenges of updating knowledge bases, clarifying prediction rationales, and correcting factual inaccuracies. By combining a pre-trained sequence-to-sequence model with a detailed context index, RAG ensures more accurate and fact-based context generation. This innovative method not only reduces the likelihood of misinformation, but also enhances the interpretability and adaptability of the model, setting a new standard for knowledge-intensive natural language processing tasks [18, 15, 19, 14].

3. Method

3.1. Retrieval-Augmented Generation (RAG)

Retrieval-Augmented Generation is an advanced method designed to enhance the factual correctness and relevance of generated content by integrating external knowledge from retrieved documents into the generation process. This approach is crucial to address the inherent limitations of LLMs, such as hallucinations and outdated knowledge, thus based the generated output on current factual information from external sources [3].

The RAG framework employs retrieval mechanisms to obtain a list of relevant documents that provide contextual knowledge to support the generation process. This integration is intended to reduce the likelihood of hallucinations and update the information with the latest research [3, 16, 11]. The typical RAG workflow includes three essential steps: corpus partitioning and vector indexing, identifying and retrieving chunks based on vector similarity, and synthesizing a response conditioned on the retrieved chunks [14].

RAG combines both the innate knowledge of LLMs and additional information fetched from external documents to generate more accurate and contextually rich responses. This dual reliance on expansive background documents and generative capabilities ensures comprehensive responses to user queries, particularly in knowledge-dense NLP tasks. As a result, RAG often outperforms conventional sequence-to-sequence (seq2seq) models and certain retrieve-and-extract architectures [16].

3.2. SentimentCareBot Architecture overview

Figure 1 illustrates the proposed model for the SentimentCareBot architecture. This model comprises a baseline selection through query translation and a vector database that simplifies retrieval through similarity search. Subsequently, a sentiment analysis ranker was used to filter the retrieved documents based on their relevance and sentiment score. Finally, the LLM uses the re-ranked documents, alongside the initial input query, to generate a final response.

3.3. Data Preparation

In this study, we used the ”Mental Health Counseling Conversations Dataset,” accessible through Hugging Face [4]. This dataset consists of 3,512 collections of questions and answers sourced from two online counseling and

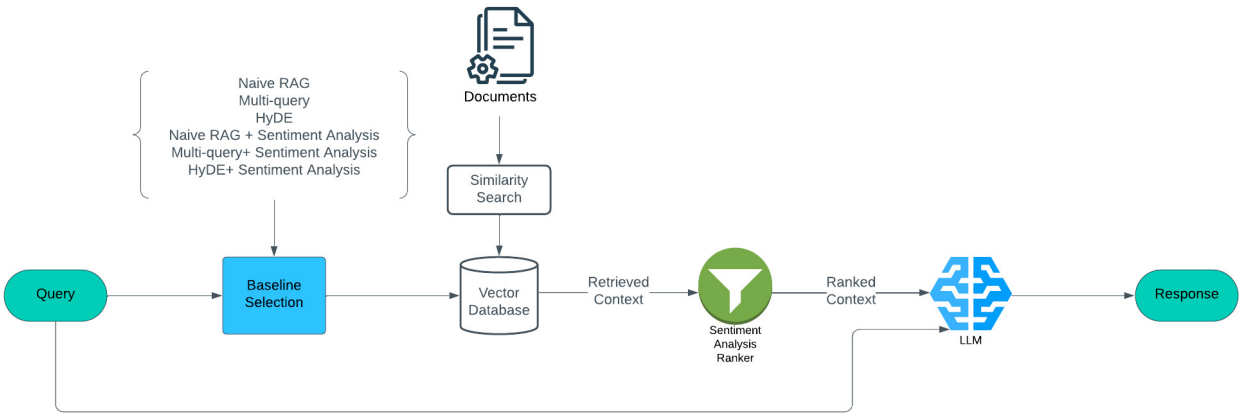


Fig. 1: SentimentCareBot Architecture

therapy platforms. The dataset provides answers to the questions that cover a wide range of mental health topics. We started by splitting the documents into chunks. This process involves dividing large documents into smaller and more manageable sections to minimize information loss. Next, we created a vector database to store and retrieve vectors. These vectors are lists of numbers that represent data within a multi-dimensional space. Using vector search methods, it is possible to locate similar data by querying for the nearest vectors in this space. Finally, a similarity search retrieves relevant documents from the database. We used Facebook AI Similarity Search (FAISS) as our similarity search metric. FAISS is a specialized library for efficient similarity search and clustering of dense vectors [9, 22].

3.4. Baseline Selection

For our baseline selection, we used Naive RAG, a conventional RAG model that simply employs the original user question for document retrieval. The Multi-query approach expands a single user query into multiple similar queries with the assistance of an LLM by generating several alternative questions that reflect the intent of the original query from different perspectives [17]. And Hypothetical Document Embeddings (HyDE), which generates a hypothetical document based on the query using LLMs [13].

3.5. Sentiment Analysis

The study analyzes the significant improvement sentiment analysis can make in the different RAG models, with the retrieval process for documents with better relevance to the question that forms the basis for the final answer. Although the documents retrieved by the FAISS similarity search may be relevant to the query, Sentiment Analysis can alter the priority of the retrieved documents by considering the highest sentiment score. We employed "cardiffnlp," a pre-trained model from Hugging Face [7], for text classification sentiment analysis. This model outputs the sentiment label and sentiment score. By applying this model, we can rank the retrieved documents based on their sentiment label, from positive sentiment to neutral sentiment to negative sentiment. After ranking the documents according to their sentiment label, those with similar sentiments are also ranked based on their highest to lowest scores.

3.6. LLM

For experimental purposes, we selected "*gpt-3.5-turbo-0125*" and "*mistral-large-latest*" due to their cost-effectiveness and ease of implementation. Given the token limitations established by the APIs provided by "*OpenAI*" and "*MistralAI*", it was critical to regulate the number of queries used for each session.

3.7. Evaluation Data and Metrics

The evaluation data consists of the complete 106 question-answer pairs, which represents approximately 3% of the entire dataset, and the remaining 97% of the dataset was used to construct the vector database. For our evaluation, we used the LLMs-as-judges metric, called the Retrieval Augmented Generation Assessment (Ragas) [12]. Ragas evaluates RAG systems and introduces a suite of metrics that do not rely solely on ground-truth human annotations. Such as the "*Faithfulness*," which assesses the factual consistency of the generated answer relative to the given context, the "*Answer Relevancy*," which assesses how pertinent the generated answer is to the given prompt, and the "*Answer Correctness*" which evaluates the accuracy of the generated answer in comparison to the ground truth that offers a comprehensive evaluation of your system's performance.

4. Results

The study evaluates a variety of advanced RAG models using metrics such as *Faithfulness*, *Answer Relevancy*, and *Answer Correctness*. A comparative analysis is presented through boxplots to visualize the distribution of these metrics on the complete 106 question-answer pairs and the subset of five question-answer pairs of the evaluation data. The ANOVA and Tukey HSD tests were then used to determine the statistical significance of the differences observed in both evaluation data. Then the token usage and latency of the subset of the evaluation data are demonstrated through a boxplot.

4.1. Comparative Analysis of Metrics

The faithfulness evaluation remained relatively stable for both the complete and subset evaluation. In terms of Answer Relevancy of the complete evaluation data, as depicted in Figure 2 show greater consistency across different techniques. The analysis for the subset evaluation data (Figure 3), the median remained consistent on different RAG models except for both Naive RAG and Multi-query using MistralAI language model where the sentiment analysis improved the median.

In terms of Answer Correctness of the complete evaluation (Figure 4) the median scores indicate a small improvements when sentiment analysis is applied across different RAG models. As for the subset evaluation as depicted in Figure 5 shows the effect of sentiment analysis across various RAG models, particularly in the Naive RAG model using the MistralAI language model. Here, the median score improved by 0.51. However, the analysis also revealed exceptions, such as in the HyDE model using OpenAI language model, where both the median and maximum scores decreased, indicating that sentiment analysis may not universally benefit all architectures. Additionally, outliers in several models indicate performance variability, which may reflect scenarios where the models either under-performed or achieved exceptionally high correctness scores.

4.2. Statistical Validation of Differences

Table 1: Tukey's HSD test results comparing different RAG models.

RAG Model	Comparison	Meandiff	P-adj	Reject
Naive RAG + OpenAI	Naive RAG + Sentiment Analysis + OpenAI	0.0287	0.9992	False
Multi-query + OpenAI	Multi-query + Sentiment Analysis + OpenAI	-0.0304	0.9987	False
Multi-query + OpenAI	Multi-query + Sentiment Analysis + Mistral	-0.1103	0.0333	True
HyDE + OpenAI	HyDE + Sentiment Analysis + OpenAI	0.0081	1	False
HyDE + Sentiment Analysis + OpenAI	Multi-query + Sentiment Analysis + Mistral	-0.1085	0.0394	True

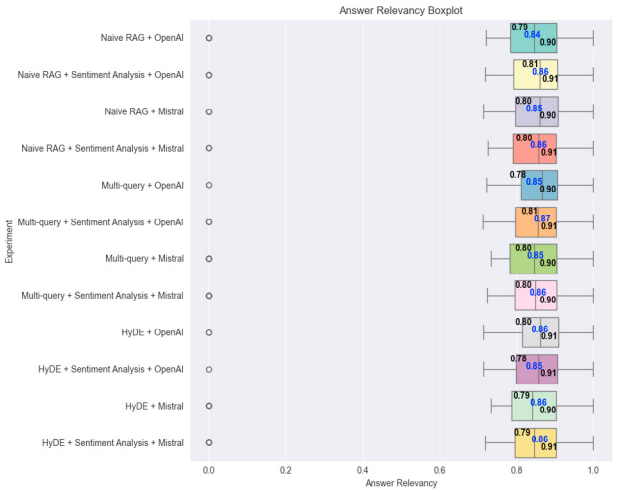


Fig. 2: Boxplot of Answer Relevancy illustrating the range distribution of Answer Relevancy scores across different RAG models.

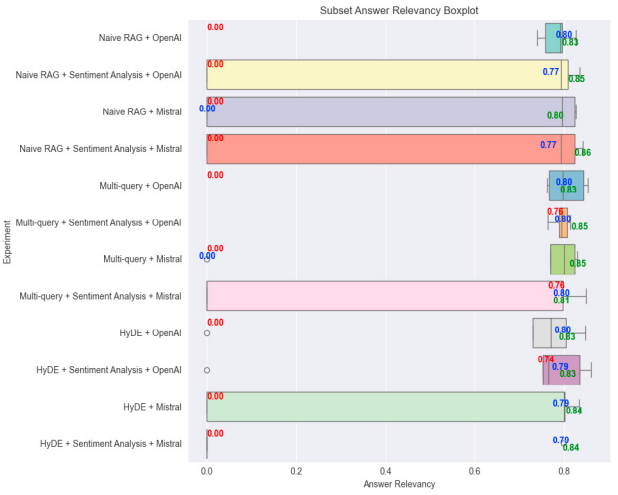


Fig. 3: Boxplot of Answer Relevancy illustrating the range distribution of Answer Relevancy scores across different RAG models on the subset evaluation data.

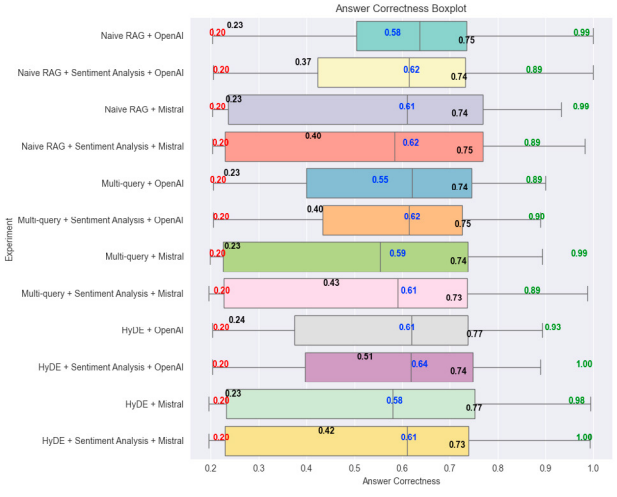


Fig. 4: Boxplot of Answer Correctness illustrating the range distribution of Answer Correctness scores across different RAG models.

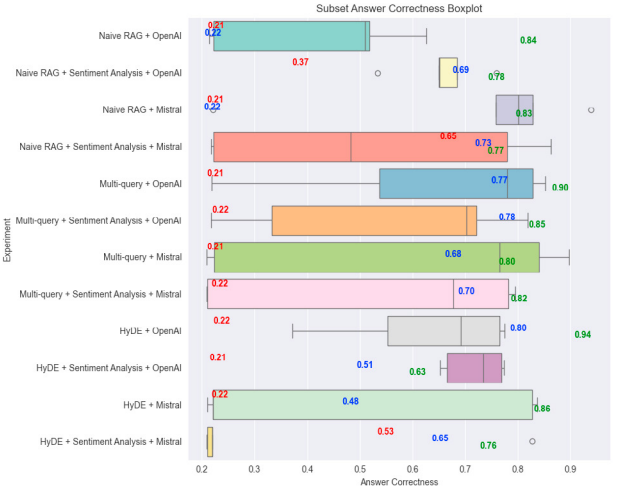


Fig. 5: Boxplot of Answer Correctness illustrating the range distribution of Answer Correctness scores across different RAG models on a subset evaluation data.

The ANOVA test we used on the evaluation results showed that there was a difference between the Answer Relevancy result groups across different RAG models of the complete evaluation data. The results of Tukey's HSD test confirmed that the Multi-query using the MistralAI language model, combined with sentiment analysis, performs better than both the Multi-query using OpenAI and the HyDE using OpenAI combined with sentiment analysis. However, other comparisons did not show a significant difference.

4.3. API performance Analysis

We evaluated the API's performance using the subset evaluation data, that consisted of five question-answer pairs. The API's token limitation constraints led us to choose this subset. The evaluation used Ragas evaluation metrics and

Langsmith[1] to examine token usage and latency for each model. By employing a manageable subset that all the RAG models could handle simultaneously, this enabled us to correctly assess the model performance.

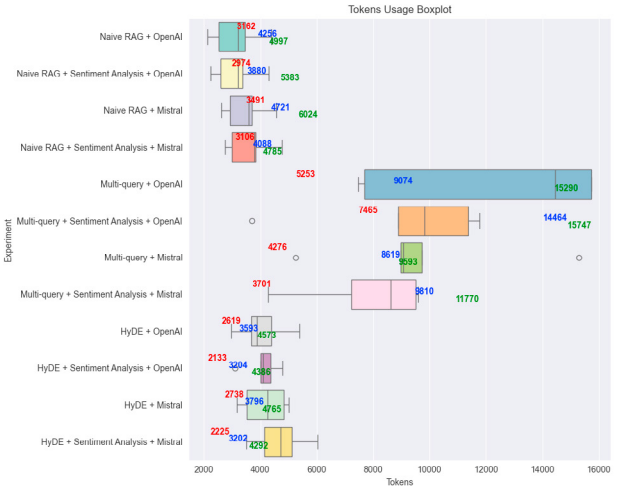


Fig. 6: Boxplot illustrating Tokens usage across different RAG models on a subset evaluation data.

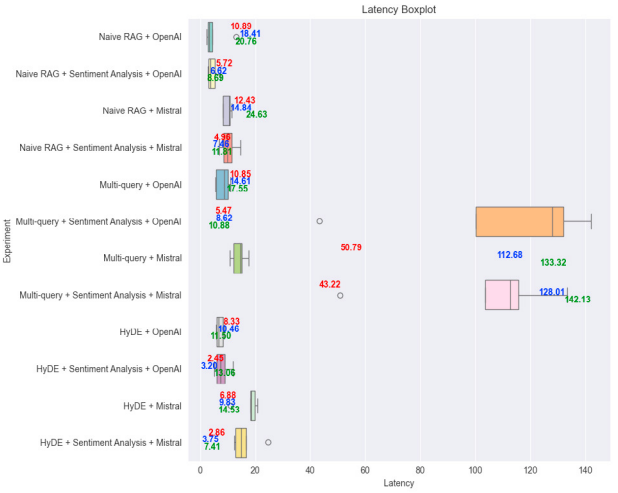


Fig. 7: Boxplot illustrating Latency across different RAG models on a subset evaluation data.

The token usage (Fig. 6) demonstrates the disparity between the performance of various RAG models. The Multi-query model using OpenAI language model demonstrated the highest token usage, with the median token count increasing when Sentiment Analysis was applied. In contrast, the HyDE baseline selection displayed the lowest token usage among the baseline selections. Despite the high token usage observed in the Multi-query baseline selection with OpenAI language model, the latency (Fig. 7) remains quite low. This indicates an efficient processing capability despite the increased token load. However, the Multi-query baseline selection using MistralAI language model displayed a higher latency, suggesting that while it may handle complex queries, it does so at the expense of response time.

5. Conclusion

In conclusion, adding sentiment analysis to the Retrieval-Augmented Generation architectures has shown a lot of promise for making mental health chatbots more useful, even though there are some challenges and restrictions associated with it. Among the tested configurations, the Multi-query approach with MistralAI's language model and sentiment analysis significantly outperforms the Multi-query method with OpenAI's language model and the HyDE approach with OpenAI's language model combined with sentiment analysis. Nevertheless, this enhanced performance comes at the cost of increased latency and token consumption, which raises concerns regarding the scalability and efficiency of API's usage. Additionally, the "*gpt-3.5-turbo-0125*" model supports a maximum of 16,384 tokens per session, while the "*mistral-large-latest*" version allows for up to 32,748 tokens. This disparity reveals critical constraints in our evaluation process, as these token limits restrict the ability to process larger datasets simultaneously. Furthermore, privacy concerns surrounding patient data contribute largely to the scarcity of publicly available mental health conversation datasets, another critical limitation. Overcoming these challenges is critical for advancing the use of sentiment analysis within RAG models in mental healthcare systems.

[1] https://smith.langchain.com

Acknowledgement

We acknowledge the support of the Natural Sciences and Engineering Research Council of Canada (NSERC) [funding reference number 06351]).

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